

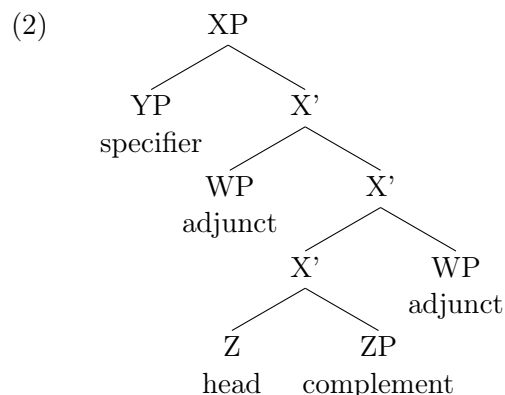
LING 102 section week 6 (3/1)

Yvette Yi-Chi Wu

1 X-bar Theory

Under the *X-bar Theory*, phrase structure rules generally follow the following schema:

- (1)
 - a. **specifier rule:** $XP \rightarrow (YP) X'$
 - b. **complement rule:** $X' \rightarrow X (ZP)$
 - c. **adjunct rule:** $X' \rightarrow X' (WP)$
 - d. **conjunction rule:** $XP \rightarrow XP \text{ conj } XP; X' \rightarrow X' \text{ conj } X'; X \rightarrow X \text{ conj } X$



Here are the rules we have so far (ignore for now constituents labelled 'spec' as placeholders):

- | | |
|--|--|
| <p>(3) core X-bar theory:</p> <ol style="list-style-type: none"> a. noun phrase: <ul style="list-style-type: none"> $NP \rightarrow (\text{spec}) N'$ $N' \rightarrow (\text{AdjP}) N' \text{ or } N' (PP)$ $N' \rightarrow N (PP) \text{ or } N (CP)$ b. verb phrase*: <ul style="list-style-type: none"> $VP \rightarrow (\text{spec}) V'$ $V' \rightarrow (\text{AdvP}) V' \text{ or } V' (\text{AdvP or PP})$ $V' \rightarrow V (NP) \text{ or } V (CP) \text{ or } V (AP)$ c. prepositional phrase: <ul style="list-style-type: none"> $PP \rightarrow (\text{spec}) P'$ $P' \rightarrow (\text{AdvP}) P' \text{ or } P' (PP)$ $P' \rightarrow P (NP)$ d. adverb/adjective phrase: <ul style="list-style-type: none"> $AP \rightarrow (\text{spec}) A'$ $A' \rightarrow (\text{Deg}) A' \text{ or } (AP) A'$ $A' \rightarrow A (PP) \text{ or } A (CP)$ | <p>(4) extended X-bar theory:</p> <ol style="list-style-type: none"> a. complementizer phrase: <ul style="list-style-type: none"> $CP \rightarrow (\text{spec}) C'$ $C' \rightarrow C TP$ b. tense phrase: <ul style="list-style-type: none"> $TP \rightarrow DP T' \text{ or } CP T'$ $T' \rightarrow T (VP)$ c. determiner phrase: <ul style="list-style-type: none"> $DP \rightarrow (DP) D'$ $D' \rightarrow D (NP)$ |
|--|--|

NOTE: sometimes the bar level X' is notated as $\bar{X}$, and the head X as X^0

* = we will defer on how to treat ditransitive verbs (e.g. *give the book to Mary*)

This X-bar structure is applicable across different languages, where according to the **Principles and Parameters** framework, 1) the *principled* behavior across languages is the content of each rule, whereas 2) the *parameter* that is specific to each language is the ordering within each rule.

Some languages have tendencies called *head-directionality*:

- (5) a. **head-initial** (e.g. English): order the head before the complement; $X' \rightarrow X \text{ ZP}$
- b. **head-final** (e.g. Japanese): order the head after the complement; $X' \rightarrow \text{ZP } X$
- c. **mixed-headed** (e.g. Mandarin): no clear preference in order of head and complement

2 Exercises

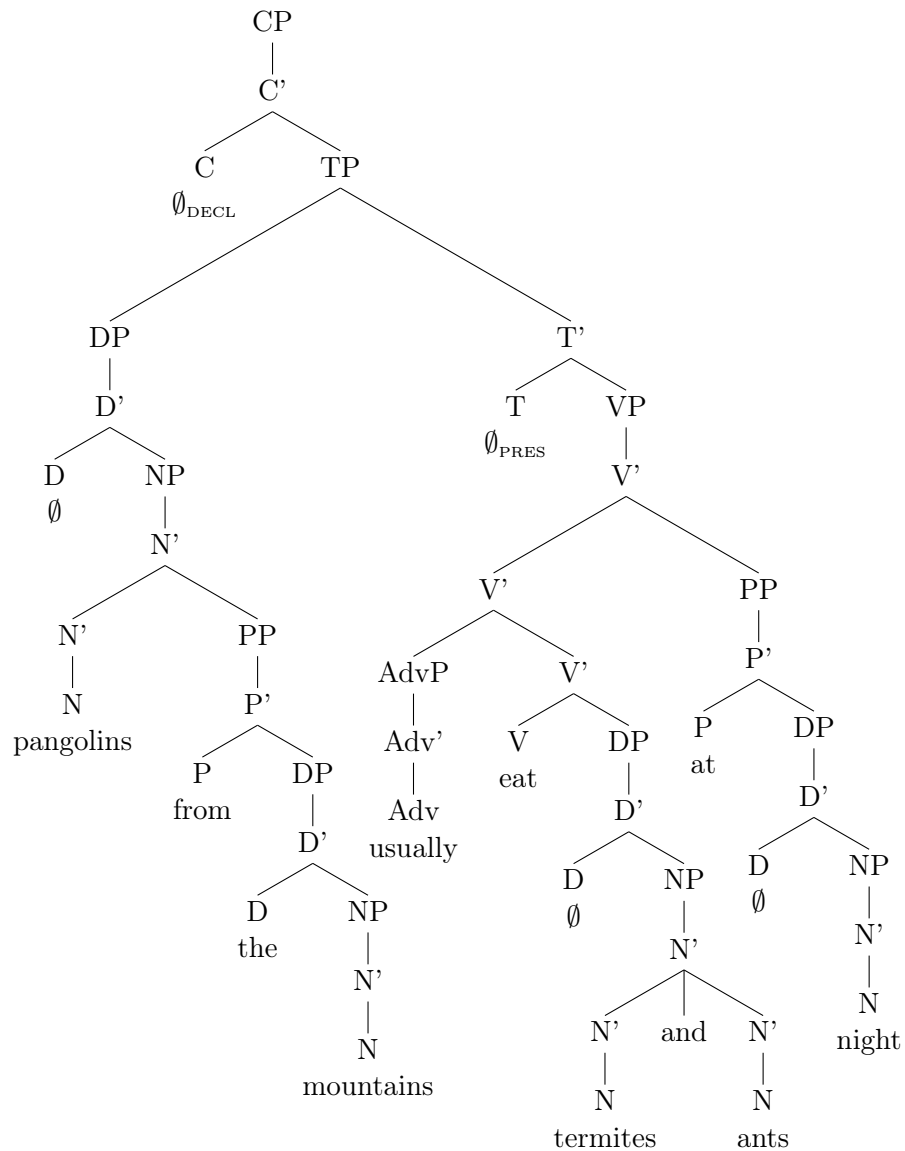
Draw trees for the following sentences:

- (6) a. Pangolins from the mountains usually eat termites and ants at night.
 - b. My favorite language is one with complicated morphology and very simple syntax.
 - c. Luna was extremely happy that she could eat her kibbles.
 - d. That it is cold outside is a very sad and miserable fact to me.
- (7) wada hpure kuxun yaku ka puqun bubu mu
PST cook like 1SG that food mother 1SG.POSS
'My mother cooked food that I like.' (psuedo-Seediq; Austronesian)

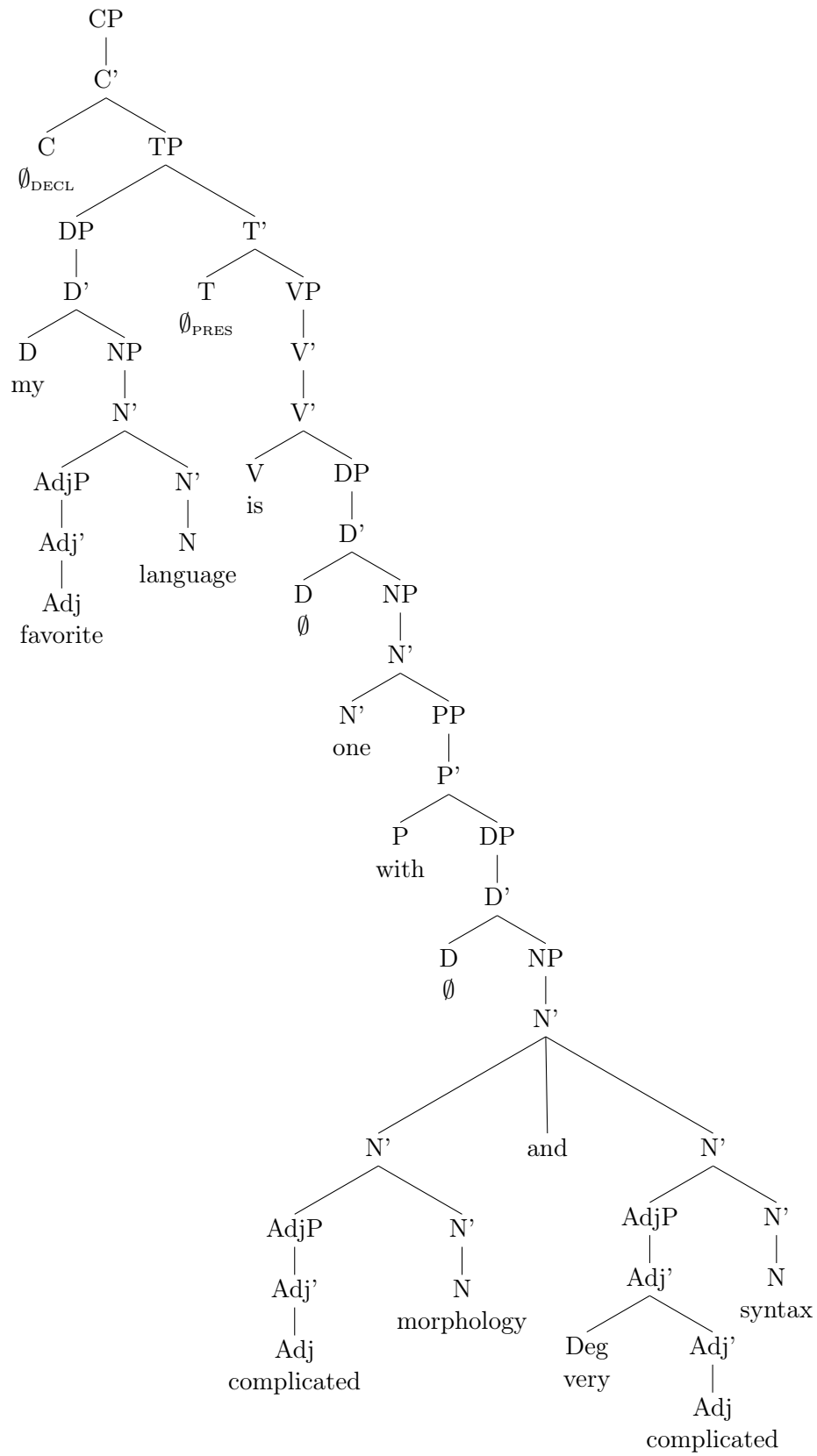
Bonus questions:

- (8) a. What do you think could occupy the specifier position of CP (i.e. Spec,CP)?
- b. Do you think CPs and TPs should also have adjunct rules? What kinds of adjuncts?
- c. Are there constituents that can be covert and not overt? Which ones?

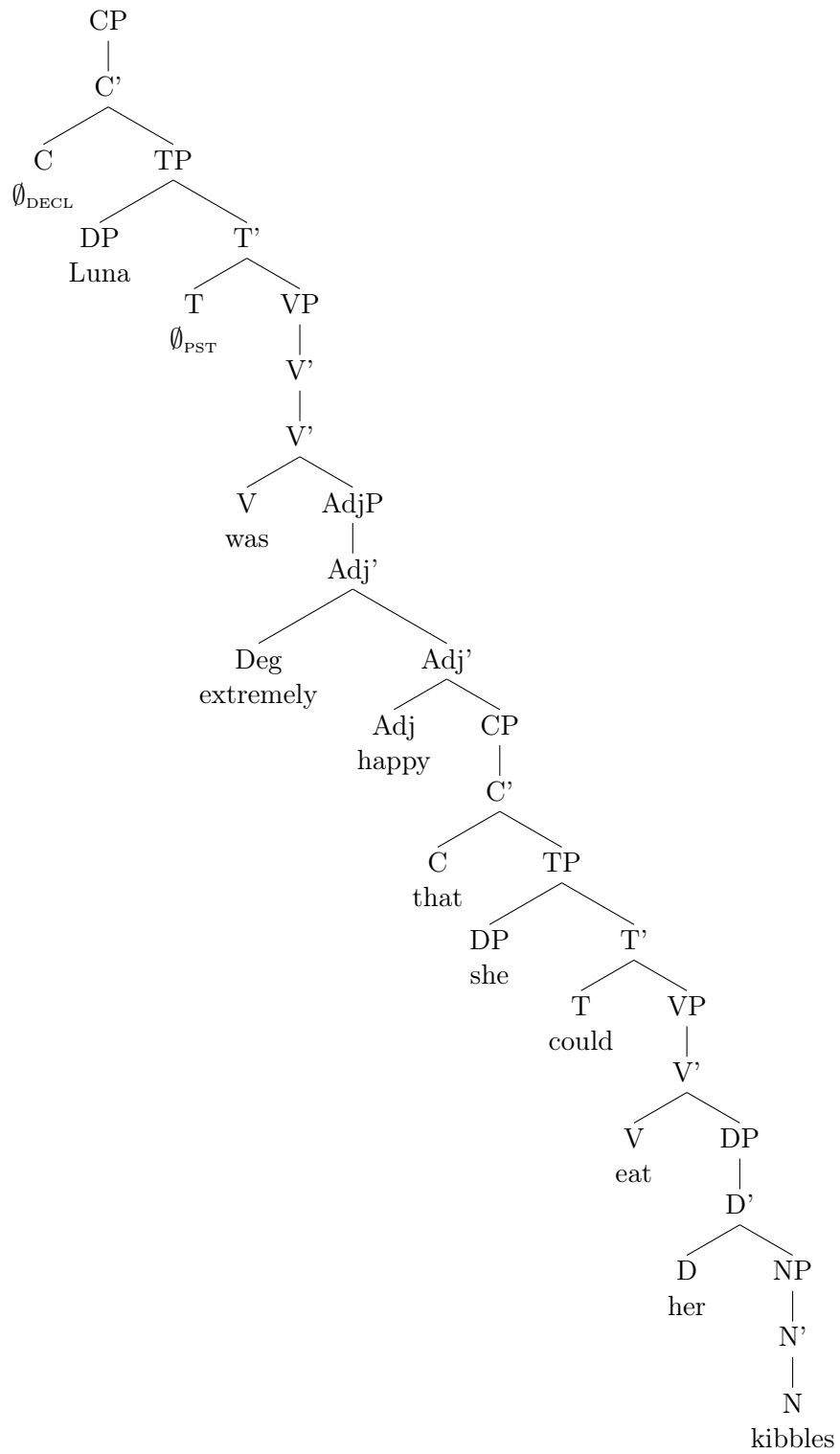
(9)



(10)



(11)



(12)

